

SILICON DREAMS

Study Guide

MODULE 3 · WEEK 3 · CM-HW-101-M3

SG-M3-08 pairs with TB-M3-08

Launch Day: Tape-out Checklist and Reflection

Run the final sign-off, submit to the ChipFoundry shuttle, prepare for 12 weeks of silence.

Learning objectives

- Execute the course-graded pre-submission checklist from end to end.
- Submit to the ChipFoundry shuttle portal without tripping any of the four rejection triggers.
- Set yourself up for a successful post-silicon bring-up in week 12.
- Write the Module 3 reflection for 300 XP.

Step 1 · The pre-submission checklist

- ☐ DRC: zero errors under SKY130A. `runs/final/reports/signoff/drc.rpt` empty of ERROR lines.
- ☐ LVS: MATCH on both Magic and Netgen. `runs/final/reports/signoff/lvs.rpt` ends with 'Circuits match: MATCH'.
- ☐ Timing: WNS ≥ 0 worst corner, WHS ≥ 0 worst corner. Both `sta-worst.rpt` slacks positive.
- ☐ Antenna: zero violations. `reports/signoff/antenna.rpt` empty of ERROR.
- ☐ LibreLane version: 2025.04 confirmed in `metrics.csv`.
- ☐ Tag: `v1.0.0-final` pushed, CI green on all three workflows (drc.yml, lvs.yml, timing.yml).
- ☐ info.yaml: fields `author` and `discord` filled in — do this LAST, right before submission.
- ☐ escape-log.md: at least one entry per module (M1, M2, M3), reflection prompts answered.
- ☐ M2 rubric: score ≥ 600 recorded. M3 eligibility is gated on M2 completion.
- ☐ Integration smoke test: all 6 tests pass (from SG-M3-02).
- ☐ Arbiter fairness: elev_share ≥ 0.02 under priority adversary (from SG-M3-03).
- ☐ AXIOM shim: all three misbehaviour tests green (from SG-M3-05).

Step 2 · Submit to the ChipFoundry shuttle

Update info.yaml with Your Real Information

```
# Update with your actual information
cat > info.yaml << 'EOF'
author: "Your Full Name"
discord: "YourDiscordHandle"
title: "Elevator Controller with Clock Gating and AXIOM Defensive Wrapper"
description: "Module 3 capstone: Elevator FSM with clock gating, AXIOM defensive
wrapper, power optimization, and silicon verification"
EOF
```

```
# Or manually edit
nano info.yaml
```

Final Submission

Now everything should be ready! The checklist should show 10/10 passed.

```
# Run final checklist
./scripts/pre_submission_checklist.sh
```

```
# Build submission tarball
./scripts/build-submission.sh
```

```
# Complete submission
./scripts/complete_submission.sh
```

```
# This produces:
#   submission/silicon-dreams-<your-discord>-m3.tar.gz
#   Contents:
#     - runs/final/final/gds/top.gds
#     - runs/final/final/lef/top.lef
#     - runs/final/metrics.csv
#     - info.yaml
#     - COURSE_HASH (sha256 of your repo at tag v1.0.0-final)
```

```
# Upload via gh CLI (preferred)
gh release create v1.0.0-final submission/silicon-dreams-<your-discord>-m3.tar.gz \
  --title "M3 tape-out submission" \
  --notes "Silicon Dreams final submission"
```

```
# Or via the ChipFoundry portal
open https://portal.chipfoundry.io/silicon-dreams/submit
# Upload the tarball. Paste your GitHub release URL.
```

Step 3 · The four rejection triggers

Missing COURSE_HASH	Portal rejects silently. Always use build-submission.sh; don't hand-roll the tarball.
LibreLane != 2025.04	Portal inspects metrics.csv. Any other version rejects.
info.yaml author blank	The course grader cross-references your Discord handle against the roster. Blank fails instantly.
GDS area > shuttle tile	Tile budget is 161 x 225 μm . If your GDS is larger, it physically cannot place. Check <code>metrics.csv</code> → <code>die_area</code> .

Step 4 · What happens after submission

The ChipFoundry grader runs a seven-stage post-submission flow on your tarball:

1. COURSE_HASH verification (1 min)
2. DRC re-run on their server (20 min)
3. LVS re-run on their server (15 min)
4. Timing re-run on their server (10 min)
5. Rubric computation (1 min)
6. Leaderboard update (1 min — only top 20 of the cohort shown)
7. Merge into the shuttle multi-project wafer (manual; happens on next shuttle close date — could be up to 4 weeks)

You will receive an email within 1 hour with your final XP score. If rejected, you get 7 days to re-submit (one re-submission allowed).

Step 5 · The 12-week wait

After the shuttle closes, fabrication takes ~10 weeks. Packaging + bring-up board shipping is another 2 weeks. During this time:

- Write up your design in `docs/writeup.md`. The course instructors will publish the best three as a cohort anthology.
- Prepare your bring-up plan — what's the first thing you'll toggle when the chip arrives? What does the oscilloscope trigger look like?
- Review the M3 textbooks once more. Chapter 8 has a bring-up checklist you'll actually use.

Step 6 · Post-silicon bring-up (week 12)

You'll receive a bring-up board with your chip soldered down. The board exposes `clk`, `rst_n`, `ui_in`, `uo_out`, and `uio` via 0.1" headers. Your first job:

8. Power the board from USB (5V, 500mA max). Confirm VDD=1.8V on the probe point.
9. Apply a 10 MHz clock (NOT 100 MHz — start slow for bring-up). rst_n low for 100 cycles.
10. Toggle ui_in[0] (request_strobe) with ui_in[4:1] = 4'h3. Watch uo_out[3:0] transition from 1 → 2 → 3.
11. If the door_open bit (uo_out[7]) pulses after ~20 cycles, you've got silicon that works. Celebrate.
12. Slowly bump the clock toward 100 MHz. Most of the cohort's chips work at 50-70 MHz in silicon even though STA reported 100 MHz — that's normal SKY130 slow-corner margin.

Step 7 · The three chip failure modes

If your chip doesn't respond on day one, narrow it to one of three:

Dead silicon	No response on any pin. Usually a power-rail short; the die was damaged in assembly. Report to ChipFoundry for replacement. Does not count against your rubric.
Partial function	Some outputs toggle, others don't. Most common pattern: arbiter grants visible on uio_out[2/3] but elevator state stuck. Check your reset synchroniser — it may have metastability at the lower clock.
Behavioural drift	Everything works but differently from simulation — off-by-one states, shifted counts. Almost always an SDC issue that snuck through. File it as a teaching moment in #m3-silicon.

Step 8 · The M3 reflection (300 XP)

In `notes/m3-reflection.md`, answer the six prompts below. Minimum 500 words total. The graders actually read these.

13. What was the hardest single moment in Module 3, and what pulled you through it?
14. If you had to do Module 3 over again, which decision would you change first?
15. Which of the five AXIOM defensive rules will you take into every future project, and why?
16. How did your mental model of timing change between SG-M3-01 and SG-M3-07?
17. Write one paragraph to yourself on silicon bring-up day. What are you going to want to remember?
18. What would you teach the Module 1 version of yourself, knowing what you know now?

The final rubric

Clean DRC/LVS/timing (final tag)	500 XP
Arbiter fairness + 50 XP bonus	250 XP
Clock gating + measurable power drop	150 XP
AXIOM shim all three tests green	250 XP
Reflection (500 words)	300 XP

Bring-up success at week 12	150 XP
First-try shuttle accept	100 XP bonus
TOTAL possible	1700 XP

The last thing to do

Push the tag. Submit the tarball. Close the laptop lid. Module 3 is done. AXIOM saw you coming — the shim held. The shuttle closes in 3 days. Your silicon will arrive. We'll see you on bring-up day.

E8873C

You built a chip. An actual chip, on an actual wafer, that actual photons will hit when you plug in the bring-up board. Whatever grade this earns, that's the accomplishment. Silicon Dreams, cohort 2026-spring, signing off.